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Tech Bytes | Asia Pacific

TurboQuant – Implications for Technology

This compression algorithm makes AI inference 8x faster while using 6x less memory. It affects only the KV cache during inference and gives much more output per GPU. The read-through is positive for hyperscalers and LLMs given the ROI opportunity. It's a long-term positive for computing and memory.

Implications for memory – neutral to positive long-term:

- **Short-term impact:** TurboQuant targets only the Key Value (KV) cache during inference (this is the temporary key/value vector that grows with context length). Models weights (HBM usage on GPU/TPU) and training workloads are not affected but it allows 4-8x longer context on the same hardware or much larger batch sizes without running out of memory. This is not a 6x reduction in memory or total hardware needed but an efficiency gain that increases throughput per GPU.
- **Long-term impact:** The Jevon's Paradox effect where efficiency increases total demand.

The inference economics are shifting: By shrinking data size and movement, TurboQuant aims to improve throughput per accelerator and lower cost per query. The biggest bottleneck in scaling AI services today is KV cache memory. If models can run with materially lower memory requirements without losing performance, the cost of serving each query drops meaningfully, resulting in more profitable AI deployment.

Thus, models that need cloud clusters can fit on local hardware, effectively lowering the barrier to deploying AI at scale. More applications become viable, more models remain active, and utilization of existing infrastructure improves.

In that sense, TurboQuant is less about incremental optimization and more about shifting the cost curve of AI deployment.

Broader tech implications – another DeepSeek moment:

- **The read-through is positive for hyperscalers and model platforms:** We cite the ROI opportunity from much cheaper per-unit quality in long-context inference and retrieval-heavy applications.
- **The implications for computing and memory are neutral near term:** Better compression means lower memory traffic and GPU-hours requirement per workload. However, a lower cost per token can also lead to higher product adoption demand (e.g., larger batch sizes, longer context). This may be negative at the margin for the software layer as compression can be embedded directly into platform infrastructure.

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Compressing the Bottleneck with TurboQuant

What is TurboQuant?

It is an algorithm designed to compress the most memory-intensive parts of AI inference, particularly key-value (KV) – a temporary memory cheat sheet so that things do not have to be re-read from scratch. This cache memory is reduced 6x, compressing 32 bits to three bits per value while preserving accuracy.

Storing extra instructions to shrink data is no longer required and overhead is reduced. It reduces not only how much memory is stored but also how much data needs to be moved between memory and computing.

This matters because 6x compression is equivalent to the same hardware handling 6x more context windows (simultaneous conversations).

Addressing the persistent scaling challenge in large language models (LLM): As context windows grow, the memory required to store KV caches expands proportionally, consuming GPU memory and slowing inference.

Google Research has introduced TurboQuant, a new compression algorithm that significantly reduces the memory and computing cost of running large AI models, with no accuracy trade-off and no retraining required. The technology targets the KV cache, a critical memory bottleneck in LLMs, and can also be applied to accelerate large-scale vector search workloads.

How does it work?

TurboQuant is a two-stage vector quantization scheme designed to jointly optimize compression efficiency and similarity accuracy.

First, it randomly rotates the vector and applies PolarQuant to capture most of the signal efficiently without the usual normalization overhead.

Then it takes the remaining error and applies one-bit Quantized Johnson-Lindenstrauss (QJL) to the residual so inner-product estimation stays unbiased and accurate.

Rather than simply compressing more aggressively, TurboQuant achieves compression with provable distortion bounds close to the information-theoretic limit, while remaining computationally efficient for online workloads.

Exhibit 1: TurboQuant's two-stage vector quantization scheme

Step 1- PolarQuant	Step 2 - QJL (1-bit Error Correction)
- Converts data vectors from standard (X,Y,X) coordinates into polar form (angle + radius) - Angle captures meaning/direction; radius captures signal strength - Eliminates the need for expensive normalisation steps -- the grid boundary is already known - Removes memory overhead inherent in traditional quantisation methods	- Uses the Johnson-Lindenstrauss Transform to shrink residual error to a single sign bit (+1 / -1) - Acts as a mathematical bias-eliminator on whatever compression error remains from Step 1 - Zero additional memory overhead -- the 1-bit correction is absorbed into the total budget - Preserves attention score accuract at 3-4 bit compression levels

Source: Google Research, Morgan Stanley Research

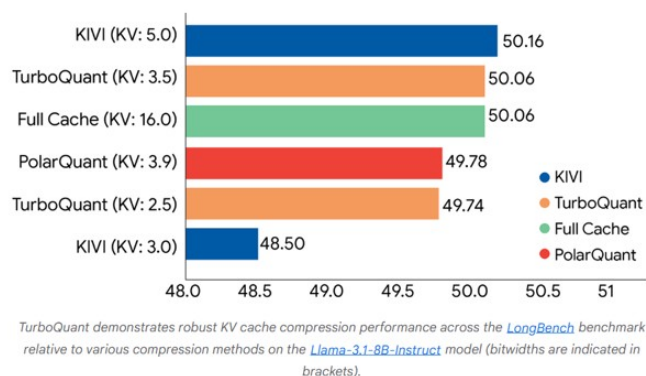
What do the results show?

- **Memory efficiency:** ≥6× reduction in KV cache memory with no measurable

accuracy loss across benchmarks including LongBench, Needle-in-a-Haystack, RULER, and ZeroSCROLLS

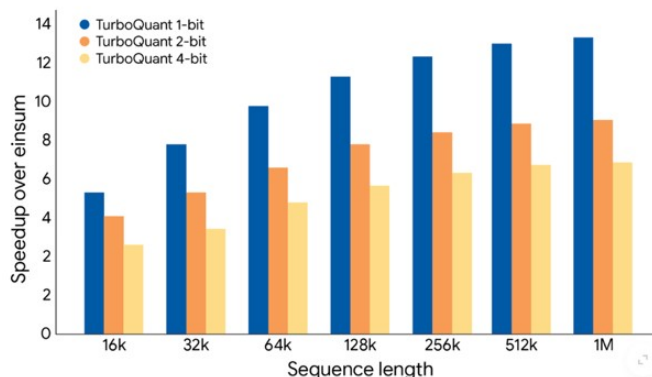
- **Compute performance:** Up to 8× speed-up in attention computation vs. FP32 baseline on NVIDIA H100 GPUs
- **Deployment simplicity:** Compression to ~3 bits without model fine-tuning, enabling drop-in integration
- **Search performance:** Outperforms prior state-of-the-art vector search methods (e.g., PQ, RabbiQ) in recall accuracy, despite using smaller codebooks and no dataset-specific tuning

Exhibit 2: TurboQuant demonstrates robust KV cache compression performance I



Source: Google Research

Exhibit 3: TurboQuant illustrates a substantial performance increase in computing attention logits within the KV cache across various bit-width levels



Source: Google Research

Why does this matter?

Structural reduction in inference cost: TurboQuant directly targets KV cache memory, one of the fastest-growing cost components in LLM inference. A 6× reduction in KV cache footprint meaningfully lowers memory requirements and improves hardware utilization, translating into lower cost per token and higher throughput per GPU. This is particularly relevant as inference, not training, becomes the dominant driver of AI workloads.

Extends effective GPU capacity: While compression reduces memory requirements, the near-term impact is less about lowering absolute GPU demand and more about unlocking higher utilization and longer context lengths on existing hardware.

Hyperscalers are likely to reinvest efficiency gains into:

1. Larger models/longer context windows;
2. Higher query volumes;
3. Improved latency SLAs.

As a result, efficiency gains may be absorbed by demand expansion, limiting downside risk to GPU volumes.

Positive for inference-heavy platforms and edge deployment: The ability to compress to ~3 bits without retraining enables drop-in deployment across existing models, lowering barriers to adoption.

This is particularly valuable for:

1. Hyperscalers scaling inference workloads,
2. Enterprises deploying private LLMs, and
3. Edge/on-device AI, where memory is constrained.

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Total	3,710		906			1636	

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